

Chapter 12

Detection and Measurement of Particulate Contaminants

Doru Florescu¹, Daniel R. Rodier², Brian A. Knollenberg²

¹*KLA-Tencor Corporation, Milpitas, CA, United States*; ²*Particle Measuring Systems, Boulder, CO, United States*

Chapter Outline

12.1 Measurements of Particles and Defects	660	Other Vacuum Equipment	667
12.1.1 Overview of Particulate Contamination and Defect Analysis	660	12.2.4 Wafer Defect and Particle Measurement Technology	668
12.1.2 Overview of Particulate Contamination and Defect Measurement Equipment	661	12.2.5 Defect Inspection Equipment	669
12.1.3 Overview of Cleaning Process Steps and Equipment Requiring Particle Monitoring	661	12.2.5.1 Unpatterned Wafer Defect Inspection	669
12.2 Defect and Particle Measurements on Wafers	662	12.2.5.2 Patterned Wafer Defect Inspection	672
12.2.1 Overview of Defect and Particle Detection on Wafers	662	12.2.5.3 Wafer Back Surface Defect Inspection	673
12.2.2 Overview of Particulate Contamination Analysis on Wafers	663	12.2.5.4 Wafer Edge Defect Inspection	674
12.2.3 Particle Monitoring on a Wafer for Cleaning Tools	664	12.2.6 Limits of Analytical Measurement Techniques	674
12.2.3.1 Particle Monitoring of Wet Stations	665	12.2.6.1 Calibration	674
12.2.3.2 Particle Monitoring of Plasma		12.2.6.2 Uncertainty and Repeatability	675
Equipment and		12.3 Particle Measurement in Liquid Chemicals	677
		12.3.1 Optical Particle Counters	677

12.3.1.1	Special Considerations for Optical Particle Counters	680		Systems and Recirculating Baths	689
12.3.1.2	Particle Counter Performance	684	12.4	Particle Measurement in Air and Gas	691
12.3.1.3	Calibration and Particle Counter Verification	687	12.4.1	Particle Measurements in Gases	691
12.3.1.4	Applications of Particle Monitoring in Liquids	687	12.4.2	Particle Measurements in Air	693
12.3.1.5	Ultrapure H ₂ O Distribution Systems: Trending and Alarming	688	12.5	Molecular Contamination Measurement	694
12.3.1.6	Chemical Distribution		12.5.1	AMC Monitoring	694
			12.6	Future of Particle Contamination Measurements	695
			12.6.1	Particle Detection on Wafers	695
			12.6.2	Particle Detection in Liquids	696
				Acknowledgments	696
				References	696

12.1 MEASUREMENTS OF PARTICLES AND DEFECTS

12.1.1 Overview of Particulate Contamination and Defect Analysis

Detection of defects on the wafer surface is a critical step during the integrated circuit (IC) manufacturing process. Nondamaging and nondestructive defect monitoring allows the wafers quickly to be checked for imperfections and then returned to the manufacturing process. This chapter focuses on the equipment used to detect and measure particulate contamination, which contributes to the imperfections that are referred to as defects, on the wafer surface. Many commercially available systems are manufactured to measure both particle size and concentration on the wafer surface, in the air, or in water and chemicals. The commercially available system for particulate contamination analysis that classify the particle size and measure the concentration fall into three broad categories:

- 1. Detection and classification of particles and other defects on a wafer surface. Presented in [Section 12.2](#).
- 2. Measurement of particle concentrations in UPW (ultrapure water, H₂O) and process chemicals. Presented in [Section 12.3](#).

3. Measurement of particle concentrations in the air and other gases. Presented in [Section 12.4](#).

In addition, airborne molecular contamination (AMC) can directly or indirectly lead to defects on the wafer surface (refer to Chapter 2). Monitoring techniques for this contaminant is presented in [Section 12.5](#).

12.1.2 Overview of Particulate Contamination and Defect Measurement Equipment

Commercially available defect measurement equipment is found in every IC manufacturing facility and is used to monitor the cleanliness of the wafer during processing or the cleanliness of the air and chemicals to which the wafer is exposed. The equipment for defect inspection on wafer surfaces is typically high-throughput automated equipment that can be integrated with other defect analysis equipment, for example, a scanning electron microscope (SEM). The equipment for particulate contamination control in UPW and chemicals is installed and integrated within these fluids' distribution system. Any deviation from the control limits signal may indicate that maintenance is required on the lines. Monitoring particles in air is typically performed at critical sites in the manufacturing area where wafers or reticles may be exposed. This equipment, strategically placed in the cleanroom or inside of process tools, can detect spikes in particulate contamination and can signal an out-of-control situation that must be corrected. The specific components that detect the particles, the theory, and the mechanism of the equipment are discussed in [Sections 12.2 and 12.3](#).

12.1.3 Overview of Cleaning Process Steps and Equipment Requiring Particle Monitoring

Essentially every processing step in the manufacturing of an IC requires monitoring the number of defects, since researchers clearly have shown that yield is related to the amount of defects and particulate contamination [1,2]. Process monitoring for the presence of particles in wet cleaning solutions, both UPW and chemicals, and in air which contribute to a decrease in IC device yield. Defects are monitored to record the level of defects as the wafer progresses through the entire process flow [3]. At some process steps, particles constitute a large percentage of the defects, as discussed later in this section. With the decrease in device feature size, smaller and smaller particles and defects must be detected. Chapter 2 outlines the relationship of defect density to yield.

In particular, particles that inadvertently are deposited on the wafer surface from the processing steps or wafer handling are monitored by single-wafer defect detection systems. In addition, the systems processing the IC devices

also are monitored to assure they are not sources of particles. Particles that originate in H₂O and chemicals are routinely monitored to assure the liquids achieve specified purity levels. Particle concentration is monitored at each chemical station. In-line real-time particle monitoring allows rapid detection of a process that is out of control. The same approach is used when monitoring the air in the cleanroom, in equipment front end modules (EFEMs), and even in front opening unified pods (FOUPs). When an out-of-control point occurs, the necessary cleaning or preventive maintenance of the UPW and chemical delivery systems, airflow system, and processing equipment can take place to eliminate the contamination source. Mitchell and Knollenberg present an overview paper of monitoring particles in situ [4].

Defect monitoring equipment is typically based on an optical light scattering technology [5] with sufficiently high sensitivity to provide detection of very small defects or particles. This includes surface scanning technology [6] used on wafer surfaces and laser-based optical particle counters (OPCs) for real-time particle monitoring of liquids and gases [7].

12.2 DEFECT AND PARTICLE MEASUREMENTS ON WAFERS

12.2.1 Overview of Defect and Particle Detection on Wafers

Defect counting metrology tools determine the following parameters:

- Number of defects
- Effective size of defect
- Location of defect
- Type of defect: particle, pit, stacking fault, scratch, etc.
- Single particles or defects, versus clusters of particles or defects
- Haze

Defect monitoring equipment inspects the front and back surfaces of the wafer, including the edge regions, top and bottom bevels, and apex. Back surface particles can cause detrimental yield issues such as creating a nonflat wafer surface during a lithography step by raising the wafer out of the focal depth of the stepper lens. The ITRS (International Technology Roadmap for Semiconductors) Front End Processing requirements state that particles less than half the size of the smallest linewidth will lead to detrimental yield decrease if they fall on the active portion of the devices [8]. That means at a 22-nm linewidth the critical particle diameter will be 11 nm at a defect density of 0.011 particles/cm² or for a 300 mm wafer, less than 38 particles. In general, as the critical dimensions of semiconductor devices continue to shrink, the size of “killer defects” also decreases as the industry moves toward 10-nm and even smaller design nodes. Current-generation particle detection is 20 nm for a blanket wafer with 0.1 nm root mean squared roughness [9].

The detection of the particle location on the wafer surface may help in the determination of the source of the contamination and may help in the recognition of trends within the wafer defectivity profile, thus helping to troubleshoot process problems that can lead to a decrease in yield. Monitoring is usually performed after thin-film deposition steps, after etching and cleaning a feature, after chemical mechanical polishing (CMP) steps, and after other critical process steps. For example, wet cleaning process steps remove unwanted contamination, including particles, thus to qualify the process for production it is imperative to have a high level of particle removal efficiency (PRE). The PRE is outlined in Chapter 4, where particles count measurements are performed before and after a cleaning step to determine how efficient the cleaning step is at removing the particles. In addition to identifying the contamination source, it is also important to accurately determine the location of particles and other defects on the wafer surface to obtain the correct count of added and removed defects [10], rather than just a simple before/after measurement, which often gives misleading information.

Back surface and edge defect detection methods are used to detect particles before process steps that are sensitive to particles in these locations, for example, particles that may have an impact on the depth of focus (DOF) at a lithography step [11] and flaking and delamination of films at the edge of the wafer [12]. Elimination of edge defects, especially the removal of CMP slurry and any particles caused by chipping of the Si wafer during the transport and handling process, is important. Thus monitoring these locations is critical for defect assessment. With the use of immersion lithography, particle monitoring is important, because the fluid introduced between the lens and the wafer transports particles across the wafer surface, and from the edge to the interior [13].

12.2.2 Overview of Particulate Contamination Analysis on Wafers

Many particle problems manifest themselves as defects on wafers and have the potential to cause yield loss [14]. Using a system to monitor defects on wafers is advantageous because the defect can be classified. Defects are classified by inspection tools based on their scattering characteristics. In addition to point defect detection, the latest inspection tools can provide full-wafer surface morphology information using the background scattering (haze) data collected from the substrate surface [15]. A more detailed explanation of haze is given in Section 12.2.5.1.

When particles are found, the root cause of the defect and the source can be traced using various analytical techniques. However, tracing the source of the particle is fraught with difficulties because the particles that are observed on wafers may be attributed to many potential sources such as the process equipment, chemical distribution system, cleanroom, and wafer handling.

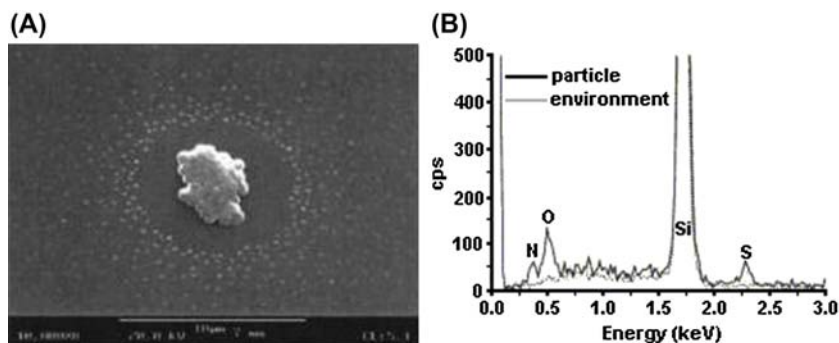


FIGURE 12.2-1 Example of particles in a wafer surface. (A) Scanning electron microscopy image of a particle on a Si wafer surface, which appears to be an agglomeration of large particles surrounded by many smaller particles. (B) Energy dispersive X-ray analysis spectra of particles indicate the elements sulfur, nitrogen, and oxygen [16]. *Used with permission from Reproduced by permission of ECS—The Electrochemical Society from N. Münter; B.O. Kolbesen, W. Storm, T. Müllerbz, J. Electrochem. Soc.*

The analysis of the particles can help in determining the source. Analytical tools such as energy dispersive X-ray analysis (EDX) can analyze the defect, which is a feature available on many SEM systems. EDX analysis is usually performed after the inspection system has found the particle's coordinates. Fig. 12.2-1 shows the analysis of a particle using EDX [16]. Chapter 13 outlines various analytical techniques for investigating the composition of a particle. Since particles can be composed of various materials, some methods may not accurately detect the composition, thus multiple analytical methods may be needed to detect the full range of elements. Also, there is a lower size limit on the particles that can be analyzed directly, depending on the spot size of the analytical tool and the haze level of the inspected wafer.

12.2.3 Particle Monitoring on a Wafer for Cleaning Tools

Successful particle control during wafer cleaning requires periodic monitoring of the performance of the cleaning equipment and processes. It is common practice to conduct particle-per-wafer-pass measurements of cleaning systems once a shift or once a day. The measured particle counts are then used as part of a statistical process control (SPC) chart to make a decision whether the process is in control. If the particle counts are outside the bounds of SPC for a certain number of measurements, corrective action can be taken. Fig. 12.2-2 shows an example of a typical particle count control chart from IC manufacturing equipment [17]. Excursions from the expected limit values are deemed out of control and corrective action is taken to bring the process back into acceptable operating limits.

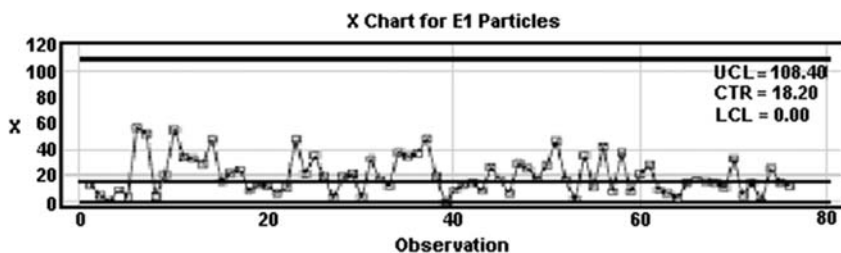


FIGURE 12.2-2 Particle count SPC chart showing control limits [17]. CTR, center control limit; LCL, lower control limit; UCL, upper control limit. Used with permission from Semiconductor International.

Monitoring systems for particulate contamination starts with a particle-grade monitor wafer. The wafer is premeasured at the desired particle size sensitivity and then processed through the manufacturing equipment to be tested, after which a final particle count is taken. In most cases, a starting wafer with a low particle count is used to determine how many added particles are deposited on the wafer surface. The wafer is first scanned before being processed to obtain a baseline particle precount. However, for process steps that remove particles, a high particle count wafer may be used—provided it is not contaminated with unwanted impurities, such as metal. The wafers are then processed through the cleaning or other equipment and a postprocess particle count is obtained. In the past, a simple delta measurement of the postcount minus the precount was used to give the number of particles added or removed from the wafer, this being referred to as per-wafer-pass measurements (PWP). Now more sophisticated techniques, such as map-to-map or defect source analysis [13,18], are used instead of PWP count. These techniques use the precise x - y locations of the defects to accurately determine the added and removed counts [17].

12.2.3.1 Particle Monitoring of Wet Stations

The particle-per-wafer-pass measurement on a wet processing station combined with routine data on chemicals, H_2O , and N_2 quality is adequate to alert an operator when the process begins to go out of control. When the particle-per-wafer-pass measurement is not in control, a particle-partitioning test can be run to isolate the problem.

Partitioning testing: testing the entire process and then segmenting the process to isolate the substep that is problematic. Particle counts should be taken using the following tests:

- Entire process: chemical tanks or process modules, rinser, and dryer
- Partition 1: dryer only
- Partition 2: rinser and dryer
- Partition 3: chemical 1 tank or module, rinser, and dryer;

- Partition 4: chemical 2 tank or module, rinser, and dryer
- Any subsequent partitioning needed

By comparing this result of the partitioning test with the particle count from the entire process, the process step responsible for the excursion can be determined. If the process bath or module is indeed the source of contamination, obtaining particle count data for the liquid chemicals is recommended. This can be done at the central distribution facility or at the point of use with a particle counter for liquids. If the baths are determined to be the cause of contamination, the following per-wafer-pass tests are recommended.

- With and without recirculation of process bath. This determines if the filter is shedding particles.
- With and without continuous drip of rinse tank nozzles for low flow. This determines if there is bacteria buildup in the nozzles.

Additional tests also can be performed to analyze the origin of the particles; some tests and the effects can be:

- Detection of high levels of total organic carbon (TOC) in the rinse tank. For hydrophobic wafers, the particle count increases with increasing TOC.
- Discoloration of tubing, bath materials, etc. This identifies if the materials of construction are shedding particles or reacting adversely with the chemicals. Metallic contamination in acid lines will gradually produce a greenish-brown color in perfluoroalkoxy pipes.
- Inadequate laminar airflow in and around wet bench process equipment or inadequate exhaust flow. Problems with the exhaust could cause cross-contamination of vapors from adjacent baths, e.g., HCl and NH₄OH vapors could react to form NH₄Cl particles.
- Poor integrity of O-rings, seals, etc. This is especially important for O₃/H₂O systems where the O₃ has been known to attack the polymeric materials of these seals.
- Specific contaminants detected. Surface analysis of the particles to determine the composition that may point to the source.

Particle monitoring techniques are used for characterizing and optimizing the performance of wet immersion systems, as well as single wafer wet processing tools. Particle monitoring is used to understand the impact of various process and equipment parameters such as filters, cleaning chemistry, drying, and wafer storage on PRE, and to develop a more effective wafer cleaning process [19].

In addition to particles, wafer haze also is used to study the PRE of cleaning tools. In a study, the researchers used haze measurements to characterize local PRE with 600 $\mu\text{m} \times 600 \mu\text{m}$ lateral resolution [20]. The local haze is used to determine the cleaning efficiency in different regions of the wafer for different process conditions and cleaning times. These experimental

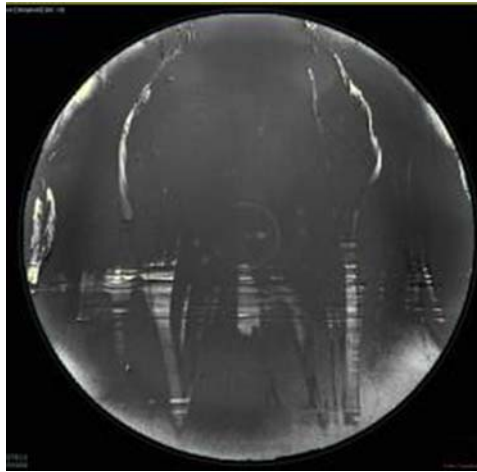


FIGURE 12.2-3 Wafer map showing wet etch residues detected in the haze map, and not in the defect channel [15]. *Used with permission from SEMI.*

data are also used to establish a quantitative relationship between the cleaning time and PRE. In addition, shallow residue (which sometimes is not detected as a defect) can also be effectively captured by the haze channel of wafer inspection systems, as can be seen from Fig. 12.2-3 [15].

12.2.3.2 Particle Monitoring of Plasma Equipment and Other Vacuum Equipment

Processing equipment such as plasma strip, etch, and other vacuum tools are routinely tested for particle excursion in much the same way wet processing equipment is tested. One concern of testing the number of particles added during plasma processing is the extra analysis needed to determine a defect on the wafer that may have arisen due to the plasma process itself. Particles in plasma systems may be caused by:

- Buildup of unwanted residue from the stripping and cleaning process on the sidewalls of the chamber.
- Inadequate venting and purging of the process chamber and loadlocks.
- Handling errors from the robot and end effector.
- Incomplete or inadequate cleaning and conditioning of the process chamber after maintenance.

Particle partitioning tests are run to isolate the source of the particulate contamination. These tests are outlined:

- Entire process: process chamber with plasma, loadlock, and handler
- Partition 1: handler only

- Partition 2: handler and loadlock
- Partition 3: process chamber without plasma, loadlock, and handler
- Any subsequent partitioning needed

In the case of multiple process chambers, isolation of the individual process chamber should be performed.

12.2.4 Wafer Defect and Particle Measurement Technology

Defect inspection equipment for monitoring cleaning processes is segmented into two types: systems that measure defects on blanket test wafers and patterned device wafers. [Section 12.2.5.1](#) outlines the blanket wafer defect detection technology and equipment and [Section 12.2.5.2](#) outlines the technology and equipment for patterned wafers.

Most fabs monitor wafers for defects at various critical steps in their process flow. Inspection systems are used to characterize device (patterned) and blanket (unpatterned) wafers for defects. Analysis of this defect data allows the IC manufacturer to increase yield by monitoring the product for defect density and determining the quality of the manufacturing process; this is typically the first step in defect root cause analyses. In addition, these data are used for assessing the manufacturing tool's ability to produce high-yielding devices. As a result, the ability to accurately and repeatedly detect, locate, size, and classify small defects on a wafer surface with these instruments is of paramount and increasing importance, as the semiconductor industry moves to smaller device sizes.

Defect inspection tools are able to give maps with coordinates and attributes defined for each defect to allow observation and classification of the type of defect, of which particles are one type [\[21\]](#). Most commercially available tools also perform grading, also called review, and classification of defects. For the purpose of simplicity, the remainder of this section will focus on unpatterned wafer inspection methodologies. The typical steps involved in defect inspection and analysis on blanket wafers are (1) detection, (2) sizing, (3) classification, and (4) analysis [\[22,23\]](#).

Detection is the use of the light scattering system to detect defects, also called light point defects (LPDs) on the wafer surface. Any scattering above a nominal threshold of light intensity signifies the presence of a defect. The method by which these LPDs are detected is discussed in [Section 12.2.5.1](#). The defect scattering information and defect location (x - y coordinates on the wafer) are stored by the system as part of the scan. The most sensitive commercially available systems in 2016 achieve 17- to 19-nm detection sensitivity on bare Si wafers. State-of-the-art systems also have an integrated wafer mapping utility to monitor surface haze variations, which can be correlated to surface roughness and other process variations.

Sizing is the use of the integrated software of the system to determine the defect size. The size of the defect is determined and segmented into a bin based upon the intensity of the light scattering signal.

Classification is the use of software analysis tools to determine the type of defect by their light scattering characteristics. Besides shape, size, and location, defect review can distinguish the following types of defects, including:

- Scratches and microscratches
- Haze
- Particles
- Missing and broken line
- Crystal oriented pits (COPs)
- Stacking faults
- Air pockets
- Voids
- Area defects and clusters

Image-based review can supply additional information about the particle, such as shape and topography, if the defect is large enough. Defect binning by type and defect classification algorithms can be used to classify defects using various attributes of defects. It has become possible to classify defects (pits, COPs, microscratches, and bumps) that cannot be removed by cleaning and defects (particle, flake, and residue) that can be removed by cleaning with very good accuracy [24].

Analysis is performed on the defect to confirm the type of defect and to analyze the composition or other physical parameters, if needed. The analytical tools have the ability to read the wafer coordinates of the defect from the saved scan results and move to the appropriate area to analyze the defect. For example, commercially available SEM/EDX systems have software with the ability to locate the defect's position on a wafer using the same coordinate map obtained on a light scattering defect detection system [25].

12.2.5 Defect Inspection Equipment

12.2.5.1 Unpatterned Wafer Defect Inspection

For defect inspection, a beam of light from a laser source is used to illuminate the wafer surface. In the presence of an imperfection, which can be a particle on the surface, surface roughness, or subsurface defect, the light is distorted and a portion of the incident light is scattered [26]. Light is reflected in directions different from that of the surface reflection. The amount of light scattered from the surface imperfections, depending on the physical and optical properties of these imperfections, is measured. Light scattering by a particle [27] is a function of its size and shape, spherical [28] or nonspherical [29,30], and material properties (e.g., refractive index) for a given incident laser wavelength. When there are no defects on a flat, perfectly smooth wafer

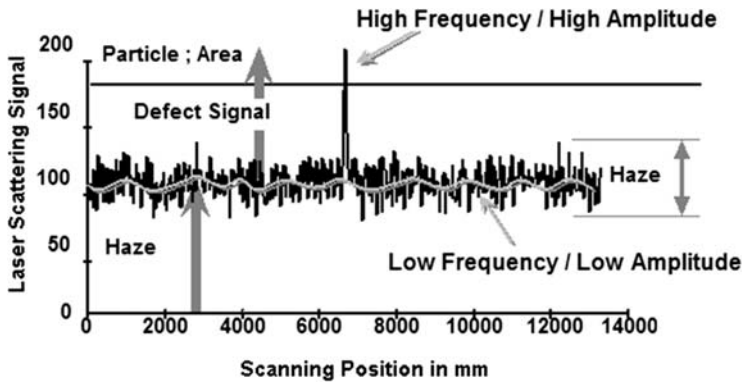


FIGURE 12.2-4 Schematic illustration of defect and haze scattering signals from a wafer surface [16]. Used with permission from SEMI.

surface and thus the angle of reflected light equals the angle of incident light, this is referred to as specular reflection; in this case all the light is reflected and none is scattered. However, all wafers scatter light because of imperfections and discontinuities on their surface. This background scattering (referred to as haze) usually determines the noise threshold that limits the smallest defect size that can be detected by the scanner. A schematic illustration of haze and defect signals and thresholds is shown in Fig. 12.2-4 [15].

As can be seen in Fig. 12.2-4, particles or other defects on the wafer surface will have a stronger scattering signal than the wafer surface. Depending on the wafer type, desired sensitivity, signal to noise ratio, and tool type, an intensity threshold is set in the scan recipe. All scattering events with intensity higher than the threshold are reported as defects, and the signal below the threshold is reported as haze. This is the basic principle of scattering-based defect inspection [26]. The ability of an inspection tool to repeatedly detect defects at a certain threshold is called its capture rate. This parameter is used to define tool specifications, to set thresholds for recipes, and for comparing scan results across different tools and scan conditions. A more detailed description of capture rate can be obtained from the SEMI M50-1101 standard.

Commercially available wafer inspection systems have both brightfield and darkfield (scattering) detection capability. In general, darkfield has better sensitivity for detecting small defects. A reference that compares them is [31]. A third technique, differential interference contrast [32], is a microscopy-based technique used primarily for detecting nonscattering defects, such as shallow residue or surface topography.

Darkfield inspection [33] detects light scattered away from the specular beam, regardless of whether the light is incident on the wafer at a normal angle, i.e., perpendicular to the wafer, or at an oblique angle, i.e., other than perpendicular [34]. A photomultiplier tube is used to collect the scattered light.

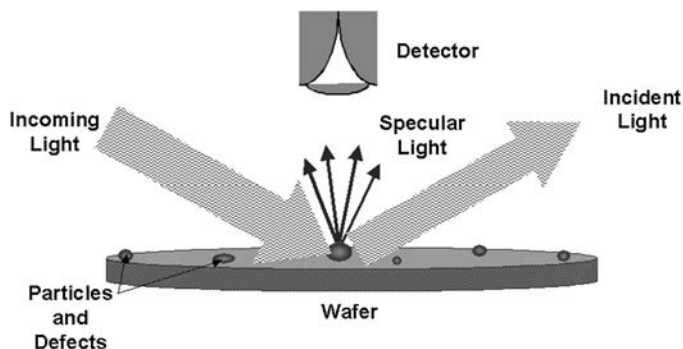


FIGURE 12.2-5 Schematic of darkfield scattering-based defect detection.

A schematic of defect detection by light scattering is shown in Fig. 12.2-5. Particles and other discrete defects are not the only cause of light scattered from a wafer [35].

Surface roughness, organic films, residue, and other surface film from unwanted contaminants all result in haze scattering, which must be distinguished from defects [36–38]. Surface roughness arises from the Si wafer not being perfectly smooth; for example, the Si surface has different planes of the Si crystalline bonding structure. Chapter 9 discusses this structure. The wafer surface can also be roughened by various surface treatments such as etching, plasma damage, and other IC fabrication processes. This surface roughness, damage, and other surface morphology changes can be characterized with subnanometer resolution by haze measurements. Controlled studies have shown good correlation between local haze and surface roughness, as measured by atomic force microscopy, on bare silicon [39], polysilicon [40], and other films. This ability of haze scattering to provide full-wafer surface roughness information becomes critical in light of the sub-4 Å surface roughness requirements predicted in the ITRS [8]. Haze scattering also has shown the ability to detect watermarks and other shallow wafer residues, which are known as yield killers [38].

Defect inspection on bare Si wafers is typically performed using light scattering tools [26,41]. Sensitivity and throughput are the primary focus of such tools designed for unpatterned wafer inspection. Sensitivity improvements in recent years have been obtained by moving from visible to ultraviolet (UV) wavelengths using laser and other deep ultraviolet (DUV) sources. The shorter-wavelength DUV light is sensitive to smaller defect sizes, with current tools showing sensitivity down to 19 nm [42]. The DUV light also has a much smaller penetration depth into Si than visible light, 10 nm, instead of several hundred nanometers. This enables the inspection of silicon-on-insulator (SOI) substrates [43], which are used in high-performance microprocessor applications. Thus the scattered light in a

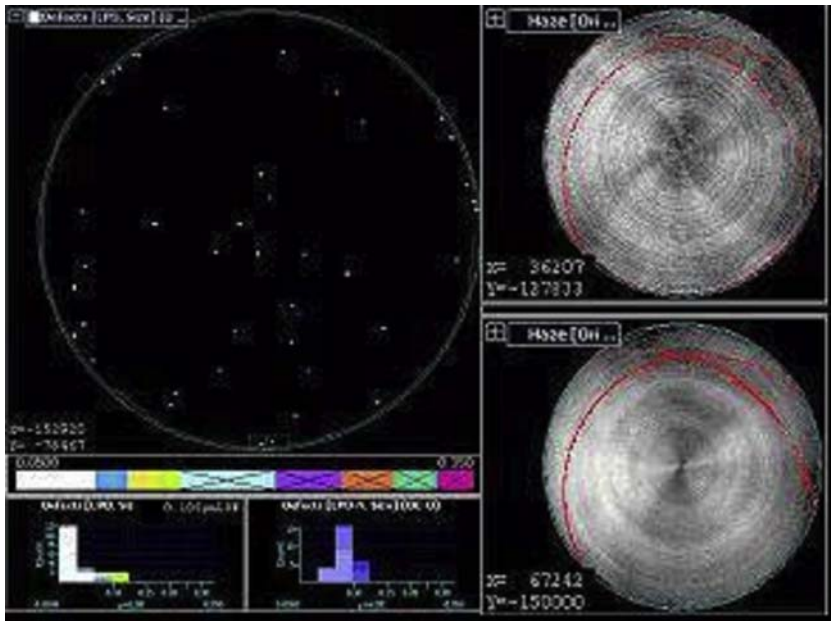


FIGURE 12.2-6 Inspection result of a silicon-on-insulator unpatterned wafer, showing particles, haze, and shallow scratches. *Used with permission from KLA-Tencor.*

UV-based tool is mostly due to surface defects and anomalies, giving more accurate and yield-relevant defect results [43]. Fig. 12.2-6 shows the inspection result on an SOI wafer, showing particles, haze, and shallow scratches detected by the haze analysis algorithm [44].

12.2.5.2 Patterned Wafer Defect Inspection

Semiconductor manufacturers use patterned wafer optical inspection systems to provide early warning of process drift, allowing engineering to take corrective action before yield is affected. The criteria for detection of defects on patterned versus unpatterned wafers are different because for patterned wafers it is important to recognize defects and imperfections that are incorporated into the device structures. The detected imperfections may be lines that have opens, shorts, or width variations, and impurities that should not be present on the wafer, such as particles, resist and implant residue, organic haze, or particulate contamination. Recognition of defects on pattern wafers requires complex imaging techniques performed with pattern recognition software and algorithms [45]. Defects on pattern wafers are detected with a method similar to that used for blanket wafers, which is discussed in Section 12.2.5.1. These optical patterned wafer inspection systems may use broadband imaging techniques, in addition to the darkfield scattering techniques described in Section 12.2.5.2. As chipmakers continue innovating by using

new materials, structures, and processes, they face an increase in new defect types, along with new noise sources that hamper defect detection.

Tunable broadband brightfield illumination technology has several advantages over a single-wavelength approach for meeting new inspection challenges and generating higher capture rates of yield-impacting defects [46]. Modeling studies as well as fab experience show that different defect types and device layers require different inspection wavelengths for reliable defect detection. A broadband source spanning DUV through visible wavelengths can be tuned to the optimal wavelength band to create maximum contrast, considering the specific optical properties of a given layer or defect type [46]. Its spectrum of wavelengths also can reduce color noise interference from underlying layers that can create nuisance defects [46]. Minimizing nuisance defects means that detected defects correlate more strongly to yield.

Commercially available broadband plasma inspection systems utilize an exclusive, high-brightness light source with optical modes that deliver increased resolution, contrast, and film layer penetration control for capture of a broader range of yield-critical defect types at the 10-nm design node and beyond and provide comprehensive inspection results for both line monitoring and engineering analysis applications.

Electron-beam (e-beam) inspection [47], compared with broadband brightlight inspection, provides distinct advantages in sensitivity for some process layers and has the unique ability to find voltage contrast defects in-line. However, slow throughput limits the e-beam inspection application to the early research and development phase or a very limited sampling plan (wafers per lot or area within the die) in production.

State-of-the-art patterned wafer inspection systems have detection limits below 10 nm, and throughputs ranging from a few wafer per hour to tens of wafers per hour [48].

12.2.5.3 Wafer Back Surface Defect Inspection

Back surface contamination, although not as immediately detrimental to the devices as defects on the front surface, nevertheless, can cause yield failure [49]. Cross-contamination of the wafer front surface from the same or adjacent wafers can result in killer defects migrating to device areas. In addition, back surface particles also cause DOF issues at lithography steps, often resulting in yield-killing hot spots. Nearly all fabrication processes can contribute to this issue. Touching the back surface and edge of the wafer in any manner, such as by robot end effectors and edge grips, wafer chucks and clamps, and FOUP and SMIF pods, can cause damage and particles to deposit on the back surface [50,51]. In addition, particles from processes such as CMP, wet cleans, and diffusion can also deposit on the wafer back surface during processing. The two typical uses of back surface inspection are for (1) monitoring process tool for back surface contamination and (2) monitoring patterned wafers on the back surface for yield-impacting defects.

12.2.5.4 Wafer Edge Defect Inspection

The edge region of a wafer is designated into five zones, the upper edge, the upper bevel, the apex, the lower bevel, and the lower edge as defined by SEMI M73-1013. Defects on the edge are important, both because edge defects can migrate onto the front or back surface, especially under liquid immersion, and edge defects can also contaminate photolithography immersion lens modules, which then require costly maintenance.

In addition to defect inspection, the location of film edge-bead removal (EBR) lines relative to wafer center and to the top of the wafer is of interest. Inspection cluster systems provide defect inspection, EBR metrology, and edge profile shape metrology: all-surface inspection and metrology at sensitivities as low as 100 nm and throughputs as high as 150 wafers per hour for 300-mm wafers.

12.2.6 Limits of Analytical Measurement Techniques

Detection of the particles on a blanket wafer must be differentiated from detection of other defects. In addition, particle detection on the wafer must distinguish the particle from the features. Examples of defects that must be distinguished from the particulate contamination are light-point crystal defects, surface roughness, and scratches, for example. As feature dimensions become smaller with each progressing node, the need to detect defects, including particles that are smaller than half the size of the feature, are deemed to be needed [8].

As particles become smaller, their detection becomes more difficult because of Rayleigh scattering [26]. Rayleigh scattering is usually used to describe system noise caused by light scattering from air molecules. Rayleigh scattering defines the exponential reduction in scattering efficiency as a function of the particle diameter. Halving the particle diameter reduces the amount of surface scattering by a factor of 64. Detection limits occur when the scattering intensity for small particles approach the level of the stray light in the system, which then reaches a limit that translates into the haze sensitivity. Current state-of-the-art commercially available technology has a limitation for detecting particles at 19 nm [42].

12.2.6.1 Calibration

Calibration of the particle monitoring system is crucial to the performance of the tool [52,53]. Methods used for calibration include using polystyrene latex (PSL) or silica spheres of known sizes, which have been classified using differential mobility analysis (DMA) [54] or secondary electron microscopy for comparison with the scanning defect monitoring tool. Ref. [52] outlines such a procedure. Custom calibration may be needed when the tool performance becomes an issue or when characterizing a specific defect of interest.

Calibration is performed at periodic maintenance checks or if the process drifts out of control during routine SPC testing. Typically, particles of known sizes are used for calibration at various defect dimensions. PSL spheres have a well-defined spherical shape and are commercially available with various calibrated sizes from approximately 20 nm to many microns in diameter with a tight standard deviation. PSL spheres are spherical, as the name implies; however, typical defects come in various shapes and scatter the light differently than the PSL spheres. Therefore, even if the PSL spheres give a tight measurement distribution, there is no guarantee that the same measurement will result for real defects. Also, the material the spheres are deposited on is important, as the reflectivity changes with different films or surface conditions. However, for production requirements this calibration method suffices.

Standards for calibration have been developed with SEMI and Sematech [55] and also by Mulholland et al. [56]. SEMI specification M53-0310 outlines the procedure for calibration of the scanning surface inspection systems (SSIS). The suggested sizes of the PSL spheres for calibration range from the largest to the smallest measurable size of interest, for example, those particles with an expected capture rate of 50%. The capture rate is defined by the probability of the SSIS capturing the signal of the PSL sphere at the operational set point. Section 12.3.1.3 discusses the importance of the particle size distribution (PSD) and symmetry of the histogram for liquid particle counting and is also applicable to particle and defect detection on a wafer. The particle size is defined by the scanner showing the detected PSL diameter producing the same signal by the detector as a same-sized defect, regardless of what the defect is.

In a typical IC manufacturing environment, particle standards are prepared and used to calibrate the system periodically. SEMI specification M58-1109 outlines the use of certified material for calibration and the use of DMA for depositing the particles, resulting in a tight size distribution. Many sources for PSL are available and there are sources for purchasing calibration standards. A wafer scan of PSL spheres of different sizes uniformly deposited for calibration is shown in Fig. 12.2-7. Note the sharp peak for the size distribution of each PSL sphere [42].

When calibrating a defect measurement system for the various size particles and defects that may be encountered, the hardware components of the systems are modified to optimize the detection of the scattered light. Multiple scattering channels and multiple detectors for use with a single-wavelength light source have been developed. Accurate scattering models are needed for these systems to achieve usable results [26].

12.2.6.2 Uncertainty and Repeatability

Uncertainty is the range of possible values within which the true value of the measurement lies. With respect to defect measurement systems, uncertainty is

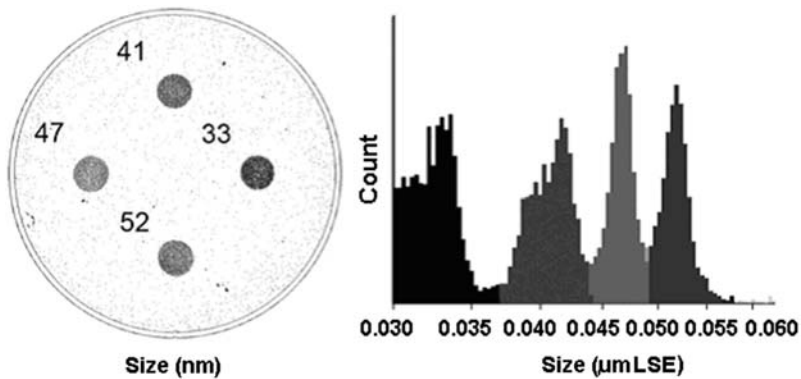


FIGURE 12.2-7 (A) Defect map and (B) histogram from a multispot polystyrene latex deposited wafer, scanned with a 30-nm threshold [42]. Used with permission from KLA-Tencor.

defined as the ability to detect the defect and its coordinates with enough precision to locate the defect again and again. This is important when performing overlay defect analysis for a patterned wafer, for example, allowing tracking of a particle throughout the manufacturing process to see if it is removed during a cleaning step.

Uncertainty is also defined as the ability of the measurement system to detect the defects that are true defects and not to detect other phenomena that are displayed as defects and also the accurate sizing of the defect as it relates to its PSL sphere equivalent. The accurate sizing of the defect is essential to the high performance standards the detection tool must meet in a manufacturing environment. Section 12.2.6.1 outlined the tool calibration for the size of the defect; however, real-time monitoring is essential to confirm this calibration. Over time the defect measurement process is tuned and optimized for the type of particles and defects encountered in production.

Repeatability is defined as the ability to take a statistically significant number of measurements and have an acceptably small variance between the measurement values to assure this value is correct [41]. The uncertainty of the measurement is expressed in the following terms:

$$CoV = \frac{\sqrt{\sigma_{RPT}^2 + \sigma_{RPD}^2}}{2(\mu_{RPT} + \mu_{RPD})} \quad (12.2-1)$$

where *CoV* is the coefficient of variance, μ_{RPT} and μ_{RPD} are the average particles counts, and σ_{RPT} and σ_{RPD} are the standard deviations; *RPT* is the repeatability test; and *RPD* is the reproducibility test. Measuring the same test wafer 10 times without unloading the wafer from the measurement station assesses repeatability, whereas measuring the same test wafer 10 times using the entire process sequence, including loading and unloading, aligning, pattern recognition, and measurement, assesses reproducibility. SPC charts are set up

to observe the long-term performance of the system and to determine acceptable limits for the repeatability; the same wafer is measured every shift or every day and plotted on the SPC chart to determine if the process is in control. Fig. 12.2-2 shows an example of an SPC chart. Statistic books such as Ref. [57] are good references for understanding SPC.

12.3 PARTICLE MEASUREMENT IN LIQUID CHEMICALS

Real-time particle monitoring of liquid chemicals is commonplace in an IC manufacturing facility. The ability to isolate the liquid chemical as the source of particles without the aid of a particle counter requires a considerable expense of time and money and, therefore, liquid particle counters are being used for high-volume production. These particle counters can quantify the level of liquid-based contamination, thus assisting in achieving better process control.

12.3.1 Optical Particle Counters

OPCs for liquids are the most popular instruments for measuring particles in UPW and process chemicals [58]. They provide a reliable, consistent, and efficient method for quantifying particle size and concentration. An OPC measures the equivalent optical size of particles based on the principal of light scattering.

A laser beam is focused on a sample cell or capillary through which a liquid is flowing. Individual particles in the liquid scatter light that is detected by a photodetector. The intensity of the scattered light is proportional to particle size, with smaller particles scattering less light. Since particle counters are calibrated with PSL standard size spheres, particles are assigned a size based on their light scattering equivalency to the PSL spheres. The calibration will be discussed in detail in Section 12.3.1.3.

Liquid particle counters can be classified as either volumetric or non-volumetric; a good overview is given by Terrell [59]. The term volumetric refers to whether or not the particle counter is viewing the entire sample flow [60]. The sample volume is the amount of fluid examined per minute during the measurement. In a volumetric particle counter the laser beam has a uniform shape across a capillary, as shown in Fig. 12.3-8. Volumetric instruments have good size resolution, provide more repeatable results because of their large sample volumes, and can accurately measure small-volume samples. When these particle counters have several channels they are referred to as spectrometers and their sizing accuracy gives them the ability to distinguish particles of different sizes. Their main drawback is that the capillary is in the viewing region and scatters some light, which reduces its sensitivity. Commercially available volumetric particle counters are limited to 0.1- μm size sensitivity.

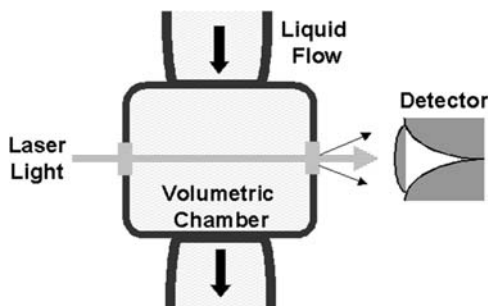


FIGURE 12.3-8 Schematic of volumetric liquid flow cell designs for particle measurements. Adapted from E. Terrell, in: M. Balazs (Ed.) 24th Proceedings of Semiconductor Pure Water and Chemicals Conference (SPWCC), Santa Clara, CA, 2005, p. 91 and R.G. Knollenberg, D.L. Veal, in: Presented at the 37th Annual Meeting of the Institute of Environmental Sciences and Technology (IEST), 1991, p. 751.

Nonvolumetric particle counters have tightly focused laser beams that provide increased sensitivity compared with volumetric particle counters, as shown in Fig. 12.3-9. These types of particle counters are often referred to as monitors because of their poor resolution and small number of channels. In addition, only a small portion of the fluid flow is examined per unit time. The sample volume can be less than 0.25% of the total flow through the instrument

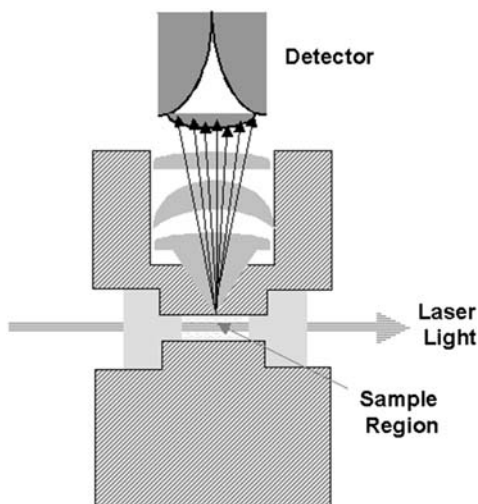


FIGURE 12.3-9 Schematic of nonvolumetric liquid flow cell designs for particle measurements. Adapted from E. Terrell, in: M. Balazs (Ed.) 24th Proceedings of Semiconductor Pure Water and Chemicals Conference (SPWCC), Santa Clara, CA, 2005, p. 91 and R.G. Knollenberg, D.L. Veal, in: Presented at the 37th Annual Meeting of the Institute of Environmental Sciences and Technology (ZEST), 1991, p. 751.

and as small as a few tenths of a milliliter per minute. High variability in the results can occur when short sample intervals are used because very little fluid is actually being examined. This is especially true when measuring UPW and very clean chemicals. The sensitivity limits for nonvolumetric counters are 20 nm (0.020 μm) in liquid chemicals and in UPW. Both types of particle counters are well suited for online monitoring, but the volumetric counters are better for off-line use with syringe or compression samplers because of their large sample volumes. An excellent review of the performance and use of OPCs is found in Knollenberg et al. [60] and in Terrell et al. [61].

A liquid particle counter technology exists [62] that uses two dimensional, digital photodetection to create images of particles moving through the laser beam. Fig. 12.3-10 shows an image of particles as they are being detected. The background light scattering from the flow cell and molecular scatter from the fluid being measured can be captured and digitally subtracted from images containing particles passing through the laser, resulting in a background subtracted image where the light scattering from particles as small as 40 nm readily can be detected (Fig. 12.3-11).

The molecular light scattering from the fluid is dispersed across thousands or millions of pixels, making the background scattered intensity per pixel much lower than a conventional photodiode detector. When a particle is detected, its image on the detector is a very small subset of all the pixels, which results in a significantly improved signal to noise ratio.

As shown in Fig. 12.3-12, a particle counter with a conventional photodetector is capable of distinguishing signals from particles relative to background light scattering from UPW. However, if the same particle counter is used to measure contamination in a fluid with high levels of molecular scatter, such as a photoresist composed of long polymer chains, it is no longer possible to detect the same-size particles because the electrical response to molecular light scattering is comparable with the response to the smallest particles the counter was designed to detect.

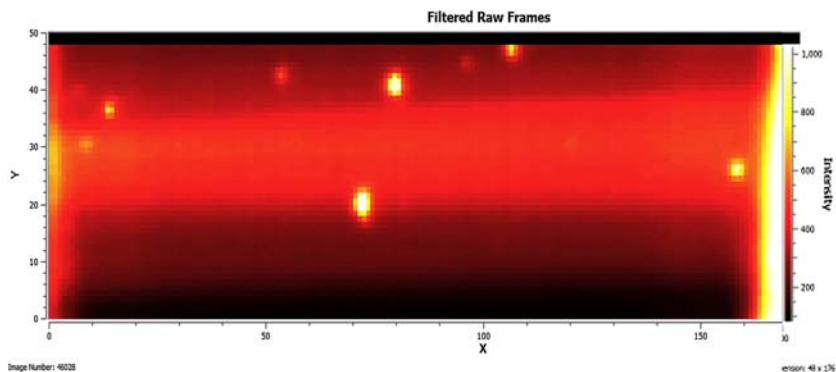


FIGURE 12.3-10 Digital detection of particles passing through the flow cell.

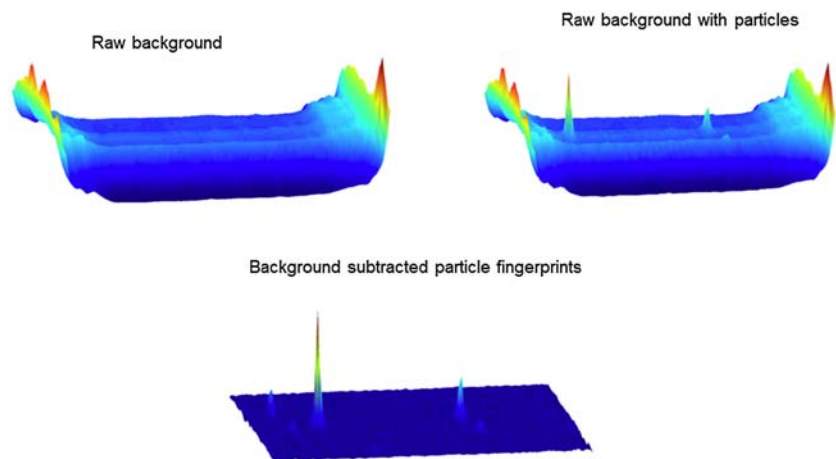


FIGURE 12.3-11 Digital processing of a particle detection event involves subtracting background scattering from a particle detection image, generating a low noise, background corrected detection image.

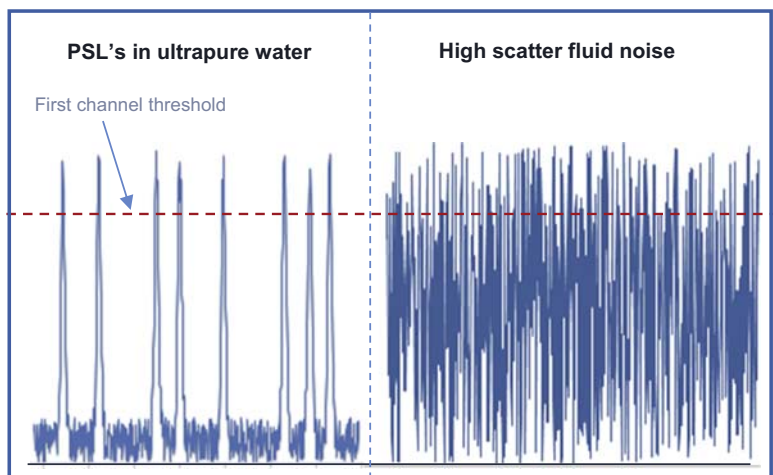


FIGURE 12.3-12 Example of electrical signals when measuring particles in ultrapure water and in a high molecular scatter fluid using a particle counter with conventional photodiode detection.

Monitoring a high molecular scatter fluid with digital detector technology allows the particle counter to be operated in high scatter mode, which greatly reduces the effect of the background scattered light, as shown in Fig. 12.3-13.

12.3.1.1 Special Considerations for Optical Particle Counters

There are special considerations that can compromise the accuracy of particle counters. These include index of refraction differences, bubbles, particle

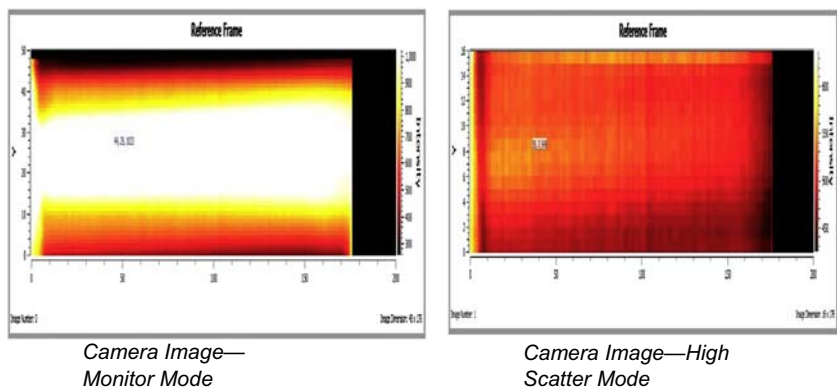


FIGURE 12.3-13 High scattering fluids can be accurately monitored by subtracting the scattered light signal via digital image processing.

counting coincidence, and contamination. For light to be scattered effectively by small particles, there must be a contrast between the index of refraction of the particle and the sample media. Index of refraction is a factor that describes how the phase velocity of light changes as it passes through a material relative to a vacuum. The complex refractive index is composed of two parts: a real component and an imaginary component. The real component describes the light scattering and the imaginary component describes the degree of light extinction. [Table 12.3-1](#) shows index of refraction values for some typical particle compositions as well as for common liquids used in semiconductor

TABLE 12.3-1 Refractive Indices of Semiconductor Materials at 780 nm

Particulate Material	Refractive Index
Silicon dioxide (SiO ₂)	1.45
Silicon (Si)	3.90
Polystyrene latex (PSL)	1.59
Sample media	
Hydrofluoric acid (HF) 50 wt%	1.29
Ammonium hydroxide (NH ₄ OH) 29 wt%	1.33
Hydrochloric acid (HCl) 37 wt%	1.41
Sulfuric acid (H ₂ SO ₄) 96 wt%	1.46
Water (H ₂ O)	1.33

wafer cleaning. Liquid OPCs are calibrated using PSL spheres in water. Particle counters size the particles by comparing the intensity of the scattered light of detected particle with the amount of light scattered by calibration spheres of different known sizes. Real-world particles may be sized somewhat larger or smaller depending on their refractive index contrast relative to PSLs in water. For example, some OPCs capable of detecting 20-nm PSLs have also demonstrated the ability to detect certain metallic particles such as gold at sizes just below 10 nm [63].

Rayleigh scattering theory [26] presents the scattering cross-section, σ_s , as:

$$\sigma_s = (2\pi^5/3) \times (d^6/\lambda^4) \times [(m^2 - 1)/(m^2 + 2)]^2 \quad (12.3-2)$$

where d is the particle diameter, λ is the wavelength of light, and m is the index of refraction contrast. The refractive index contrast:

$$m = n_{\text{particle}}/n_{\text{media}} \quad (12.3-3)$$

where n is the refractive index. From Eq. (12.3-2), the Rayleigh scattering intensity is proportional to d^6 and λ^{-4} . Also, as m approaches 1, the Rayleigh scattering cross-section and thus the Rayleigh scattered intensity approach zero.

Bubbles are an issue with liquid particle counting because they are counted as particles that can skew the actual number of particles [64]. Bubbles have an index of refraction contrast with liquid chemicals ($n_{\text{gas}} = 1.00$) making them appear as particles. Many process chemicals used in semiconductor wafer cleaning (descriptions are found in Chapter 4), easily form bubbles, including H_2O_2 , SC-1, SC-2, and hot UPW. The effect of bubbles can sometimes be reduced with the use of degasification membrane filters [65]. These, however, may be difficult to flush clean because of their high surface area and may generate particles as they age. Wettability characteristics for the membrane are important to consider because improper membrane selection can result in it becoming a nucleation site for bubble formation. If online particle counting occurs under sufficient pressure, gases can remain in solution and may not be a problem. Off-line samples can benefit from the use of a compression sampler. Compression sampling forces the bubbles back into solution before measuring the sample. If a liquid sample is collected for off-line analysis, air can be entrained during the collection, especially for more viscous fluids. These types of generated bubbles can be difficult to remove even with compression sampling. Proper sample handling practices and time may be the only solution to this specific problem. In the case of a volumetric particle counter, bubbles can be easily identified because they typically occur in the larger size channels. Fig. 12.3-14 shows the benefit of using a compression sampler at pressures of 45 psi (pounds per square inch). In this example, a pressure of 30 psi is insufficient and the bubbles are counted as 2- μm size particles.

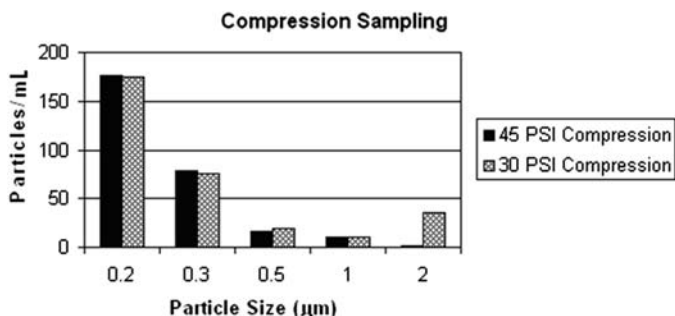


FIGURE 12.3-14 Example of particle size with different compression pressures. Compression is used to return bubbles back into solution so that they will not be incorrectly counted as particles. *Used with permission of Particle Measuring Systems.*

Particle counting or optical coincidence occurs when more than one particle is in the viewing region at the same time. In this case, multiple smaller particles get sized as one single larger particle. All particle counters have a specified maximum concentration that is design dependent. Above this limit, sizing and overall counts will be inaccurate. Fig. 12.3-15 shows how doubling the particle concentration in sequential steps eventually leads to a reduction of counts in the most sensitive channel when compared with the next largest size channel. In this case, tap water with a high particle concentration is introduced at different rates. The particle counter's maximum concentration (10,000 particles/mL) is exceeded at 1 mL/2 min. The skewed distribution provides evidence that counting coincidence is occurring. Counting coincidence can occur when the particle concentration is very high and two particles pass through the laser beam at the same time and are counted as one large particle, instead of two

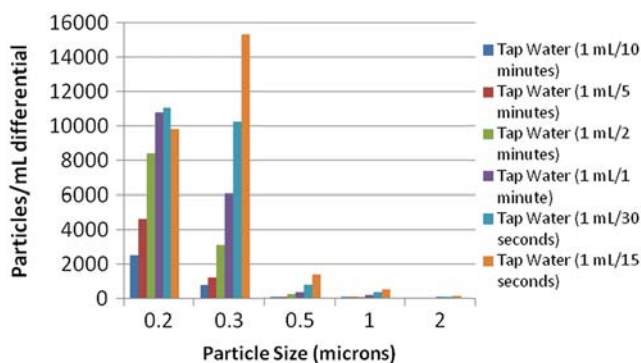


FIGURE 12.3-15 Tap H₂O with high particle concentrations introduced at various rates. Very fast introduction rates, in this case 1 mL/15 s, led to optical coincidence and reduced differential counts in the most sensitive channel (0.2 μm) when compared with the second channel (0.3 μm). *Used with permission of Particle Measuring Systems.*

smaller particles. In extreme situations, total cumulative particle concentrations will decrease with increasing particle concentrations.

Contamination can impact the accuracy of particle counter results, for volumetric particle counters, in particular. Capillaries are particularly susceptible to contamination, which will cause more light to be scattered. This results in an increase in the direct current (DC) component of the voltage signal (DC light level). It can also cause unusually high counts in the most sensitive channel. Sample cell contamination usually is not an issue with the high sensitivity nonvolumetric counters, because the viewing region does not include the optical interfaces. However, high-sensitivity instruments are affected more by small particles present in the sampling lines and valves. It is not unusual, when installing a high-sensitivity particle counter, that “clean up” times to baseline levels can last for several hours for very clean UPW systems. Changes in flow rates will affect the face velocity along the tubing and can cause particle shedding. Valves are especially prone to trapping and shedding particles and should always be fully opened or closed. Partially open valves may generate cavitation and continually shed particles.

12.3.1.2 Particle Counter Performance

Ultrapure H₂O must have very low levels of particulate contamination, outlined in SEMI specification F63-0211. A modern UPW system will typically have less than 1 particle/mL greater than 20 nm in diameter [66]. High-purity chemicals manufactured for the semiconductor industry often have at least one to two orders of magnitude higher levels of contamination than UPW, which is considered very clean. The relative infrequency of particle detection by particle counters can result in poor repeatability. The most important criteria in establishing particle counter performance are the number of particles counted per unit time. The two design criteria that directly impact the frequency of particle detection are sensitivity and sample volume.

Size sensitivity is important since there are a greater number of small-diameter particles when compared with the number of large-diameter particles. Filters for liquid systems act primarily as sieves, removing the large particles preferentially. However, they are not perfect and some small percentage of large particles will still be detected. PSDs in continuously filtered UPW systems and in chemical distribution systems have been well documented by Knollenberg [60] and Mitchell [67]. These PSDs closely approximate a power function distribution where the cumulative concentrations for different particle sizes have a relationship of diameter⁻³. The graphical representation of this is best shown using a log–log plot as depicted in Fig. 12.3-16. The rule of thumb is that when the diameter is halved there will be eight times as many cumulative particles in the liquid.

In optical particle design, sensitivity comes with a trade-off with respect to the sample volume. Since the sample volume represents the amount of fluid

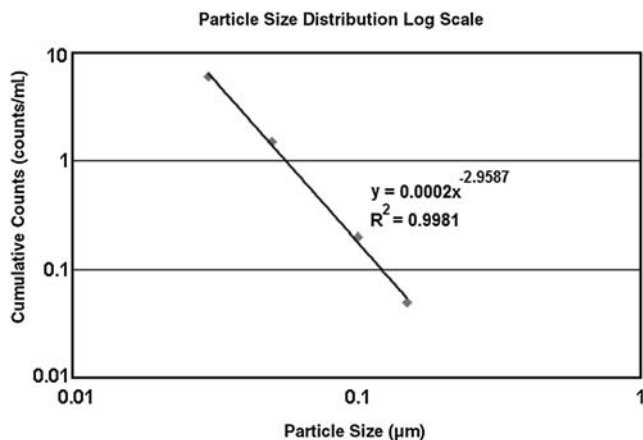


FIGURE 12.3-16 Cumulative particle count distribution in a continuously filtered liquid system [59]. Used with permission of Semiconductor Pure Water and Chemical Conference. Copyright 2005 SPWCC.

being examined per minute, a larger sample volume will result in more particles being detected. Since the number of particles detected is a function of both sensitivity and sample volume, both of these criteria affect particle counter performance; the more particles detected the higher the confidence in the result. A reasonable measure of statistical significance requires the detection of a minimum of 20 particles. Fig. 12.3-17 shows the time required for several current commercially available instruments to detect 20 particles in a fluid with a concentration of 0.2 particles/mL $> 0.1 \mu\text{m}$ with a PSD of diameter⁻³.

The precision of the particle count measurement is primarily a function of counting statistics. The standard deviation, SD , and relative standard deviation, RSD , can be calculated by knowing the total number, N , of particles counted during the sample interval [68]:

$$SD = \sqrt{N} \quad (12.3-4)$$

$$RSD = \sqrt{N}/N \quad (12.3-5)$$

When larger volumes of fluid are examined, more particles are detected during the sample interval. This reduces the sample-to-sample variation, as shown in Fig. 12.3-18, and the relative standard deviation is smaller. In addition, by knowing the number of particles counted per sample interval, the upper control limits can be calculated using probability tables. For the example shown, two particle counters with the same sensitivity ($0.05 \mu\text{m}$), monitoring the same fluid have very different control limits. This is due to the fact that Counter 1 has a sample volume 15 times larger than Counter 2 and the control

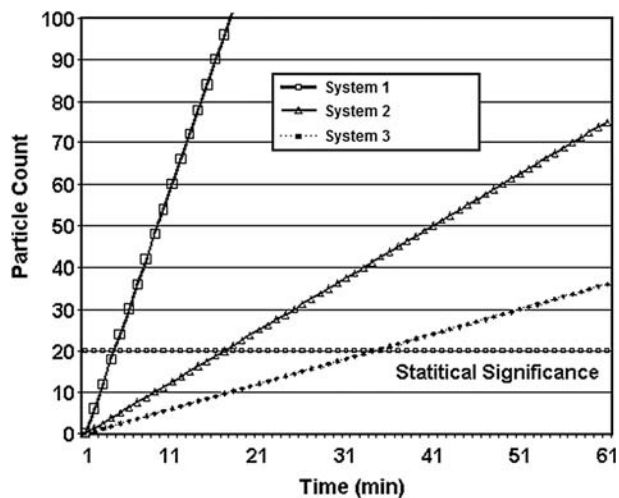


FIGURE 12.3-17 Minimum time required for accurate results require detecting at least 20 particles. The minimum of 20 counts gives a reasonable measure of statistical significance. Used with permission from Particle Measuring Systems.

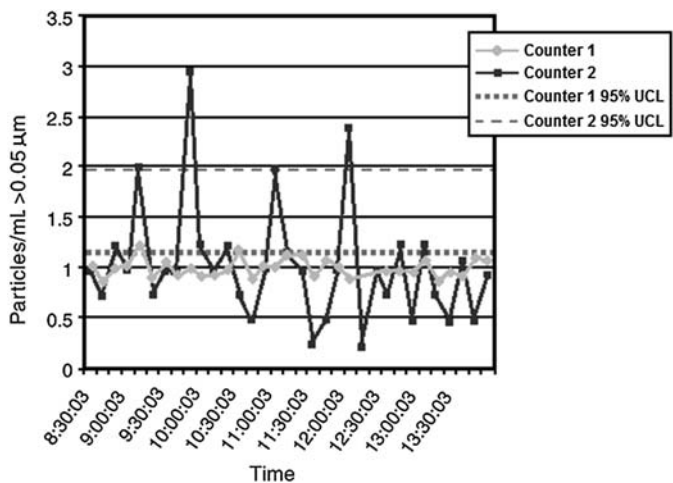


FIGURE 12.3-18 Particle counts versus sampling over a time period. The particle counter with the larger sample volume has less data variability and allows tighter control limits to be established [59,61]. Used with permission of Semiconductor Pure Water and Chemical Conference.

limits are normalized to these volumes. In Eq. (12.6), the value of 1.96 comes from a Student's t table of probability constants for the 95% upper control limit.

$$95\% \text{ UCL} = [(\text{expected counts}) + 1.96 \times SD]/\text{sample volume} \quad (12.3-6)$$

A more detailed discussion of the impact of sizing sensitivity and sample volume on particle performance can be found in Terrell [61] and Mitchell [67].

12.3.1.3 Calibration and Particle Counter Verification

Particle counter calibration, using PSL spheres as standards, establishes the particle counter's sensitivity and sizing accuracy [69,70]. Volumetric particle counters count 100% of the particles in the flow coming into the instrument; therefore if the sizing accuracy is correct, the counting accuracy is correct. Commercially available PSL spheres have a normal distribution of particle sizes around the mean particle size. As a result of this distribution of particle sizes around the mean size, 50% of the particles are above the mean size and 50% of the particles are below the mean size. The sensitivity threshold is set at the mean particle size during calibration so that the particle counter will count 50% of the PSL spheres present at the size standard being calibrated and thus all particles greater than the mean size will be counted. Therefore, calibration best can be performed by comparison with a reference instrument or with the use of count standards, which are commercially available PSL sphere solutions of a certified concentration. Refer to [Section 12.2.6.1](#) for standardization of PSL spheres.

Setting the sensitivity limit at 50% counting efficiency is important because it sets the sensitivity limit independent of the resolution of the particle counter. If values other than 50% were used, then the resolution would affect the sensitivity limit as shown in [Fig. 12.3-19](#), resulting in inaccuracies when counting real-world particles. This example shows the effect of setting the threshold incorrectly at 90% of the total PSL spheres counts. Sizing accuracy at sizes greater than the sensitivity limit can be verified for volumetric particle counters by measuring the distribution of a PSL sphere size standard [67,68]. This requires setting the channel's size threshold to be exactly the same as the PSL sphere size being tested. A cumulative distribution function is used to compute the sizing accuracy based on the percentage of PSL spheres above this threshold and the particle counter resolution. Sizing errors are typically less than 5% at the calibration particle sizes.

Nonvolumetric monitors have poor resolution and inherently poor sizing accuracy. These types of counters are calibrated by setting the sensitivity limits and their sample volumes to match a reference unit or to meet counting efficiency specifications with count standards. Channels above the sensitivity limit are set by adjusting voltage thresholds such that the distribution of the PSL sphere size standards meet predefined calibration criteria. Performance verification of a nonvolumetric monitor requires a reference unit or count standard.

12.3.1.4 Applications of Particle Monitoring in Liquids

Particle counters can be used in-line, with a syringe sampler or with a compression sampler. Particle counters installed on pressurized delivery lines

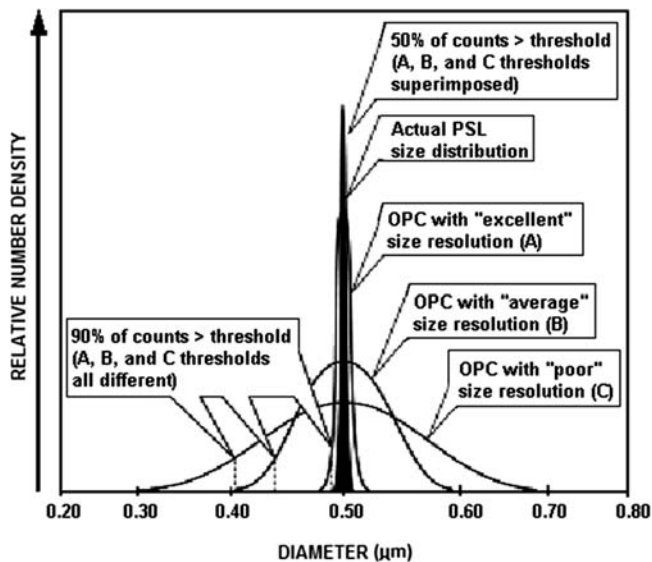


FIGURE 12.3-19 Example of particle size distributions for polystyrene latex sphere standards. The sensitivity limit of a volumetric particle counter is set at 50% counting efficiency. This ensures that the sensitivity limit is independent of resolution and that all particles greater than the sensitivity limits are counted. *Used with permission from Particle Measuring Systems.*

provide continuous sampling and are less susceptible to contamination. Syringe samplers pull a small and precise amount of fluid through the particle counter. These are useful for batch-type operation with samples of limited volume or with expensive chemicals. Finally, compression samplers are available to minimize false counts due to bubbles when measuring chemicals off-line. Compression samplers are also useful for making continuous measurements directly from a recirculating chemical or UPW bath.

12.3.1.5 Ultrapure H₂O Distribution Systems: Trending and Alarming

Ultrapure H₂O is the cleanest fluid in the fab and is critical for the manufacturing of high-yielding semiconductor devices, which is used extensively in aqueous cleaning and rinsing baths as well as in the chemical makeup of etching, stripping, and cleaning solutions. Owing to the very low concentrations of particles in UPW distribution systems, continuous in-line monitoring is normally used. Wali et al. showed the relationship between particles in UPW and random yield loss [2].

Continuous in-line monitoring minimizes the potential for introducing particulate contamination and provides high levels of statistical confidence in the results. Particles in wafer can be composed of various materials, all of which must be detected by the particle counter [71]. Although portable carts

with particle monitors are still used for troubleshooting, these have become less useful except for identifying extreme out-of-spec conditions. Owing to the extreme cleanliness of modern UPW systems, nonvolumetric particle monitors exclusively are used for contamination monitoring. Detection sensitivities in the range of 20–50 nm are commonly used.

Hot UPW can form bubbles that may be counted as particles. The best approach to eliminating this possibility is the use of a small chiller or a long length of tubing before the particle counter. The chiller should be qualified before use to ensure that it does not shed particles.

12.3.1.6 Chemical Distribution Systems and Recirculating Baths

Particle monitoring of chemical distribution systems is similar to UPW monitoring. Most particle counters are installed in-line to reduce the possibility of contamination and to provide continuous monitoring of the liquid chemicals. Particulate contamination levels tend to be somewhat higher for chemicals than for UPW. Particle counters typically monitor the incoming chemical quality, polishing loops (if used), and the distribution system to the point of process. Most chemistries exhibit more molecular scattering than H₂O, which raises the background voltage signal. This often requires an increase in the counter's voltage threshold setting for the smallest particle size channel and a reduced sensitivity specification.

Specialized mobile carts for particle counting with scrubbed exhaust and leak detection are employed to a limited extent for troubleshooting. Mobile carts suffer from the same particle issues that occur with their UPW counterparts in that long clean up times may be required and are generally only useful for identifying large excursions.

Monitoring of recirculating chemical baths can be accomplished with an in-line counter in the recirculation loop. However, it is usually preferable, depending on the processing speed, to directly monitor the bath, so that the particle counter measures the same fluid that comes in contact with the wafer. A compression sampler operating in a continuous mode is best for accomplishing this goal. When a wafer is introduced in a bath, an increase in particle concentrations is observed with the clean up to baseline levels depending on the filter recirculation rate, filtration efficiency, and rate of clean chemical addition, as shown in Fig. 12.3-20. Rapid processing can result in an incomplete return to the baseline levels, increased bath contamination, and a decrease in product quality. Some compression samplers are equipped to accept external triggers so that the sampling can be halted during the cleaning and replenishment of the bath. Gromala and Beal [72] describe parameter optimization for compression sampling to improve the data reliability.

The ITRS [8] shows the concentration for all aqueous and solvent cleaning chemicals at below 7.5 particles/mL greater than 20 nm in 2017. If the most sensitive particle counter available is limited in measurement capability to >0.03 or >0.04 μm particle sizes, it is necessary to estimate the concentration

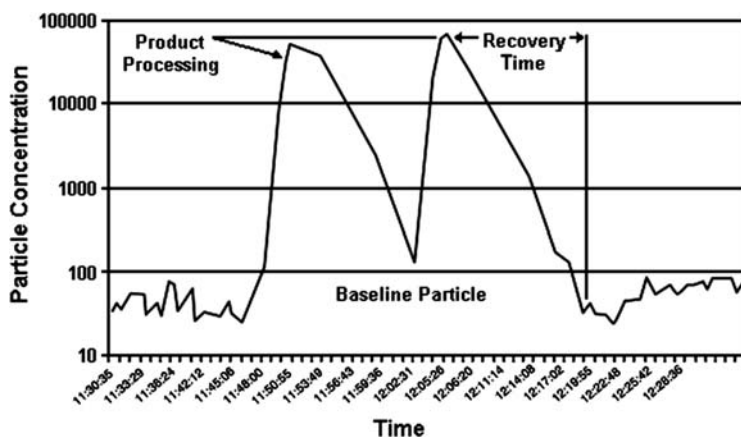


FIGURE 12.3-20 Example of particle concentration with time showing that spikes in particle concentration occurs when wafers are introduced into a bath. The particle cleanup is dependent on the recirculation rate, filter efficiency, and rate of clean chemical additions. *Used with permission from Particle Measuring Systems.*

to the smaller particle size, 20 nm, using an expected PSD of diameter⁻³. This is becoming increasingly difficult to do accurately, because real PSDs have been shown to be highly variable depending on particle filter performance and chemical distribution system design [73]. Monitors with larger sample volumes count more particles per unit time and provide somewhat better confidence in the results.

12.3.2 Nonvolatile Residue Monitor

There are several nonvolatile colloidal species in UPW that deposit onto the surface of wafers during the rinsing and drying steps of a cleaning process. These include colloidal silica, bacteria, particles, and certain ionic impurities. Blackford et al. [74–76] describe a nonvolatile residue monitor (NRM) that quantitatively measures the amount of these nonvolatile species in H₂O. Fig. 12.3-21 depicts a schematic diagram of this monitor [74]. A sample of H₂O is atomized into an air stream that carries it through a diffusion dryer. The H₂O is evaporated, leaving a particulate phase that represents the nonvolatile components. These particles are typically well below 0.1 μm in size, and conventional aerosol OPCs cannot be used to size and count them. Blackford et al. [70] recommend the use of a condensation nucleus counter (CNC) to count these particles. In a CNC, small particles are coated with a film of alcohol (e.g., butanol) that increases the effective size to ~10 μm, allowing the small-size particles to be counted by standard optical techniques. This technique can also be used to measure the nonvolatile residue of H₂O-soluble solvents by injecting pulses of the solvent into the UPW passing through the monitor [76].

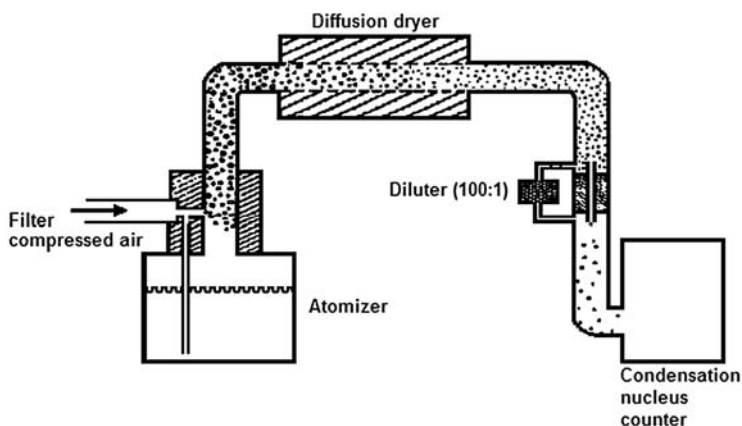


FIGURE 12.3-21 Nonvolatile residue monitor [74]. Copyright 1987 by IEST. Used by permission. Reprinted with permission from *J. Environ. Sci.* (847) (1987) 981-0100, published by IEST. www.iest.org. From D.B. Blackford, K.J. Belling, G.J. Sem, A new method for measuring nonvolatile residue for ultrapure solvents, *J. Environ. Sci.* 30 (4) (1987) 43.

It should be noted that NRM is not a particle counter and does not attempt to distinguish between real particles and the other nonvolatile materials. Some nonvolatile chemicals, such as silica, may exist as a colloid, a single particle, or dissolved in solution. In addition, its form can change with changes in the pH. Nonvolatile residue monitoring results are reported in mass or parts per billion (ppb) and are calibrated using NaCl, which is well behaved during evaporation, in contrast to real samples that experience some loss during the drying process.

12.4 PARTICLE MEASUREMENT IN AIR AND GAS

Particle counters that directly measure particles in air and gas are important tools for understanding exposure risk of wafers, reticles, and other sensitive components. Knowing the number of particles that are in the wafer environment is critical for the successful production of semiconductor devices. Table 12.4-2 outlines the cleanliness classifications for clean manufacturing environments [77].

12.4.1 Particle Measurements in Gases

Gases for the semiconductor industry are manufactured to have purity levels. SEMI specification C3-0317 outlined the purity specification for gases. Gases with high levels of AMCs, impurity gases, and particles are detrimental to IC processing. Process and purge gases can be monitored for particles in the exhaust stream and at the point of use. Particles can be generated in the gas line from various sources such as valves, regulators, and piping.

High-efficiency filters are used to remove them. However, not all particles can be removed and in some cases the filter can introduce contamination. If the filter experiences rapid changes in pressure or flow rate or mechanical failure, particle counters can be used to detect the excursion. Qualification of a gas distribution system is accomplished using a gas-phase OPC or a condensation particle counter.

There are commercially available particle counters that are capable of processing the different gases that may be used in a cleanroom, both for inert gases (e.g., He, N₂, CDA, Ar) and for reactive gases (e.g., H₂, O₂). However, corrosive gases (e.g., Cl₂, HCl) and highly reactive gases (e.g., HF) cannot be measured.

The particle size sensitivity is 0.1 μm at a flow rate of 0.1 SCFM (standard cubic feet per minute) for gases at under 150 psi [78]. When analyzing reactive gases the probe is contained in a cylindrical pressure vessel, providing a containment housing for safety, which is pressurized with N₂ during operation.

Direct sampling of the gases at their delivery pressure is preferred, using a mass flow controller (MFC) to provide a constant flow rate and the same volume for each sample. This technique works best for gases at pressures between 40 and 150 psi. Owing to the different physical properties of the gases being tested, the ability to accurately size the particles is dependent on the pressure and the composition of the gas. Particle counters are designed to account for these variables, including sensing the flow rate and the exact atmospheric pressure. Typically, a gas constant entered into the instrument's data system provides correction factors for different gases and allows the MFC to increase/decrease the flow rate based upon the chemistry of the gas [78].

12.4.2 Particle Measurements in Air

Termed aerosol particle counters, these systems measure particles in air. This technology can be used to monitor wafer and reticle exposure to airborne particles in EFEMs, in the processing equipment atmospheric pressure staging area, and in equipment storage areas, such as FOUP and reticle stockers. In addition, monitoring can be performed at various locations within the fabrication area and are useful for constant monitoring of the real-time health of the cleanroom. Cleanroom standards classify the cleanroom, as shown in Table 12.4-2. Typically, 0.1- μm particles are monitored according to the values set by the International Organization for Standardization (ISO) [77].

Handheld and stand-alone particle counters are also commercially available. Many companies make aerosol particle counters and their use goes beyond semiconductor manufacturing to include pharmaceutical manufacturing, medical apparatus manufacturing, and food processing areas, for example.

The techniques used to measure particles in gases are similar to those used for measuring particles in liquid media, as discussed in Section 12.3.1.

The particle size measured is dependent on the wavelength of light, light will not be scattered for a particle smaller than the wavelength used for the detection.

12.5 MOLECULAR CONTAMINATION MEASUREMENT

12.5.1 AMC Monitoring

AMC is gas-phase contamination that can react with or adsorb onto critical surfaces. Chapter 2 provides an in-depth discussion of AMC. A high-level comparison between molecular contamination and particulate contamination in an ISO Class 1 environment shows that there is roughly 10^{12} times more molecules of contamination per cubic meter than there are particles of contamination. This translates into 10,000–20,000 times more mass of AMC than particles per cubic meter [77,79]. The primary concern with AMC is this mass can react directly with or create a thin film covering a sensitive surface. These interactions with highly engineered wafer and reticle surfaces can cause yield loss by changing the electrical, optical, chemical, or physical properties of the surface [80]. Some examples of this are metal thin film corrosion, hazing of optics, changes in surface wettability, T-topping of photoresist, thin film adhesion problems, and uneven oxide growth.

AMC monitoring can be performed on-line or off-line [81]. Real-time, on-line monitoring generally provides the most complete and accurate data sets and the ability to rapidly detect changes in AMC levels. Off-line or laboratory techniques can often provide more specific identification of the various types of AMC that are present, but are labor intensive and have poor time resolution [82]. Another option is noncontinuous, real-time monitoring using a sequential manifold system [83]. The manifold system can be centrally located and sample tubing routed throughout the cleanroom, bringing the air sample to the analysis equipment. This allows a single set of analyzers to monitor tens of different locations, lowering the cost per sample point. However, the measurements are more likely to have issues with response time and tubing surface effects.

When selecting AMC-monitoring locations, it is important to consider the relative risk associated with each location. Locations are often classified as critical or noncritical. Critical monitoring locations require continuous real-time data at sub-ppb concentrations for contamination event detection, whereas noncritical locations usually require only periodic data at low ppb levels for trending purposes. In general, it is important to monitor where sensitive products or processes are located and near potential AMC sources in the cleanroom. If a process or area is important enough to require chemical filtration or purge gas, it is important enough to require AMC monitoring to verify contamination levels.

12.6 FUTURE OF PARTICLE CONTAMINATION MEASUREMENTS

Detection of smaller and smaller particles challenges the technological limits of traditional light-scattering techniques [84], both for counting nanoparticles in liquids, air, and on wafers. The challenges include problems due to the intensity of scattered light from the particles relative to the scattered light of the measurement cell for liquids and air, and by the wafer itself. In these cases, the particle's signal is lost in the background noise.

12.6.1 Particle Detection on Wafers

To address these challenges for wafer inspection, researchers have proposed various methods to circumvent existing problems. These include alternative light sources for darkfield and brightfield illumination for wafer inspection [85]; an amplification technique for detecting the low-level particle signal based on the interferometric amplification [9]; and image denoising by replacing the laser source with an incoherent light-emitting diode to remove the speckle noise [86]. Other improvements include evolution of patterned wafers inspection software where die-database inspection can capture large areas to stitch multiple dies together, creating a composite image [87]; extremely accurate and stable wafer stage positioning [88]; and optimized software that controls scanning speed [89].

Electron-beam defect detection systems, which suffer from low throughput [87], but can image very small defects of particles, are evolving. Multiple massively parallel e-beams have been proposed [88–90], to prevent interference from one beam to another. Researchers have found that ultra-high image acquisition rates scale with the number of electron beams [89]. UV wavelengths are still being considered for smaller particles. Researchers have shown that by reducing the inspection wavelength from deep-UV to vacuum UV and extreme UV, defect detection can be extended to smaller particle sizes [91].

Other approaches to optimize particle detection include illuminating the sample from numerous azimuth angles, to provide a lower detection limit verified by an analysis of the speckle signal that originates from the surface roughness of the substrate [92]. The multiple angles used simultaneously detect the particle or defect, whereas the individual angles provide information on the size and shape of the particle or defect.

Coating a particle on a wafer surface in a fluid, which can be removed later, then analyzing the wafer for the particles is an interesting alternative to direct on-wafer analysis. Researchers propose particles less than 10 nm could be detected [93].

12.6.2 Particle Detection in Liquids

Reliable detection of particles <20 nm [94] for liquid chemical and water is required to meet the demands of high-volume manufacturing of 10-nm devices. UPW particle size has shrunk to 10 nm for the most advanced semiconductor processes, which is beyond the capability of most some UPW treatment technologies [95]. Thus as UPW production evolves, measuring particles in water and chemical must evolve.

Not only are small particle sizes required to be removed, but also these particles are at a low concentration in the UPW [96]. Besides detection, the researchers found that filters are required to retain 10-nm and smaller particles, in their study primarily composed of silica, with carbon also present [96].

ACKNOWLEDGMENTS

The authors of the Handbook's second edition were Ed Terrell and Karen Reinhardt. The third edition authors wish to thank Mark Keefer, Paul Horn, and Kurt Haller for their contributions to this chapter. Thanks to Lisa Cheung and Uday Mahajan for contributions and John Stover for his helpful review of the chapter manuscript for the Handbook's second edition.

REFERENCES

- [1] P. Bichebois, Workshop on Defect and Fault-Tolerance in VLSI Systems, IEEE, Boston, Massachusetts, 1996, p. 124.
- [2] F. Wali, D.M. Knotter, A. Mud, F.G. Kuper, *Microelectron. Eng.* 86 (2) (2009) 140.
- [3] K. Mori, N. Nguyen, D. Keeton, R. Burns, in: B. Vasquez, H. Kawasaki (Eds.), *SPIE Microelectronics Manufacturability, Yield, Reliability*, vol. 2334, 1994, p. 62.
- [4] J.R. Mitchell, B.A. Knollenberg, *Semicond. Int.* 19 (10) (September 1996) 6.
- [5] H.C. Van de Hulst, *Light Scattering by Small Particles*, Dover Publications, New York, 1981.
- [6] M. Miyoshi, in: T. Hattori (Ed.), *Ultraclean Surface Processing of Silicon Wafer*, Springer-Verlag, Berlin, 1998.
- [7] D.L. Black, M.Q. McQuay, M.P. Bonin, *Prog. Energy Combust. Sci.* 22 (3) (1996) 267.
- [8] The International Technology Roadmap for Semiconductors (ITRS), Semiconductor Industry Association, Austin, TX, 2015.
- [9] W.D. Koek, E.J. van Zwet, H. Sadeghian, in: *SPIE Interferometry XVIII*, vol. 9960, 2016, p. 0E.
- [10] T.J. Zazado, J.R. Bottin, in: *IEEE/SEMI Advanced Semiconductor Manufacturing Conference and Workshop (ASMC)*, 2005, p. 100.
- [11] A. Carlson, T. Le, J. Hallen, *Solid State Technol.* 50 (6) (2007) 43.
- [12] J.D. Morillo, T. Houghton, J.M. Baue, R. Smith, R. Shay, in: *IEEE/SEMI Advanced Semiconductor Manufacturing Conference and Workshop (ASMC)*, 2005, p. 49.
- [13] K. Nakano, S. Owa, I. Malik, T. Yamamoto, S. Nag, in: *SPIE Optical Microlithography XIX*, vol. 6154, 2006, p. 4J.
- [14] B. Tullis, in: R.P. Donovan (Ed.), *Particle Control for Semiconductor Manufacturing*, Marcel Dekker, New York, 1990, p. 359.

- [15] T. Braun, A. Wocko, in: Sematech ISMI Symposium on Manufacturing Effectiveness, 2005.
- [16] N. Münter, B.O. Kolbesen, W. Storm, T. Müllerbz, J. Electrochem. Soc. 150 (2003) G192.
- [17] W.A. Levinson, F. Stensney, R. Webb, R. Glahn, Semicond. Int. 24 (12) (October 2001) 83.
- [18] T.Y. Teo, W.L. Goh, V.S.K. Lim, L.S. Leong, T.Y. Tse, L. Chan, J. Vac. Sci. Technol. B22 (1) (2004) 65.
- [19] K. Sano, A. Izumi, A. Eitoku, J. Snow, K. Wostyn, E. Holsteyns, P. Mertens, A. Steinbach, in: Sematech Surface Preparation and Cleaning Conference, 2006.
- [20] T. Janssens, F. Holsteyns, K. Kenis, S. Arnauts, T. Bearda, K. Wostyn, G. Simpson, A. Steinbach, P. Mertens, Solid State Phenom. 134 (2006) 233.
- [21] P.B. Chou, A.R. Rao, M.C. Sturzenbecker, F.Y. Wu, V.H. Brecher, Mach. Vis. Appl. 9 (4) (1997) 201.
- [22] C.H. Wang, W. Kuo, H. Bensmail, IIE Trans. 38 (12) (2006) 1059.
- [23] S.S. Gleason, K.W. Tobin, T.P. Karnowski, F. Lakhani, in: SPIE Metrology, Inspection, and Process Control for Microlithography XII, vol. 3332, 1998, p. 232.
- [24] M.E. Fossey, J.C. Stover, L.D. Clementi, US Patent 6,118,525 (2000).
- [25] C.R. Brundle, Y.S. Uritsky, in: A. Diebold (Ed.), Handbook of Semiconductor Metrology, Marcel Decker, N. Y., 2001, p. 547.
- [26] J.C. Stover, Optical Scattering, second ed., SPIE Press, Bellingham, Washington, 1995.
- [27] J.C. Stover, V. Ivakhnenko, Y.A. Eremin, in: SPIE Optical Metrology Roadmap for the Semiconductor, Optical, Data Storage Industries II, vol. 4449, 2001, p. 131.
- [28] P.A. Bobbert, J. Vlieger, Physica 137A (1986) 209.
- [29] T.A. Germer, in: T. Wriedt (Ed.), 7th Conference on Electromagnetic and Light Scattering by Non-spherical Particles: Theory, Measurements, Applications, 2003, p. 93. Bremen, Germany.
- [30] T.A. Germer, Opt. Lett. 27 (2002) 1159.
- [31] K.W. Tobin, in: V. Sankaran, C.M. Weber, K.W. Tobin (Eds.), Webster's Encyclopedia of Electrical and Electronic Engineering, vol. 10, Wiley & Sons, New York, 1999, p. 242.
- [32] R.D. Allen, G.B. David, G. Nomarski, Z. Wiss. Mikrosk. 69 (1969) 193.
- [33] T. Watanabe, T. Jingu, M. Noguchi, T. Hosoe, Hitachi Rev. 52 (3) (2003) 153.
- [34] W. Zhou, D. Hart, R. Roy, in: SPIE Illumination Optics II, vol. 8170, 2011, p. 0p-1.
- [35] C.R. Olson, D.M. Yalamanchili, R.W. Shorn, L. Mule' Stagno, Future Fab Intl., vol. 14, 2003. Section 6.
- [36] N. Munter, B.O. Kolbesen, W. Storm, T. Muller, Solid State Phenom. 76–77 (2001) 115.
- [37] L.W. Shive, R.E. Blank, K.H. Lamb, MICRO 19 (3) (March 2001) 59.
- [38] J.J. Shen, L.M. Cook, K.G. Pierce, S.B. Loncki, J. Electrochem. Soc. 143 (6) (1996) 2068.
- [39] D.K. Chen, K. Sun, M. Yeh, C.C. Chen, in: Yield Management Symposium (YMS), KLA-Tencor, Taiwan, 2006.
- [40] D.K. Chen, A. Steinbach, M. Yeh, C.C. Chen, S.C. Chen, M.S. Liang, in: IEEE/SEMI International Symposium on Semiconductor Manufacturing (ISSM), 2006.
- [41] P.-F. Huang, Y.S. Uritsky, C.R. Brundle, in: A. Diebold (Ed.), Handbook of Semiconductor Metrology, Marcel Decker, New York, 2001, p. 515.
- [42] D. Shorn, L. Cheung, Solid State Phenom. 103–104 (2005) 133.
- [43] C. Maleville, C. Moulin, D. Delprat, W. McMillan, J. Payne, K. Birdwell, R. Moirin, in: G.K. Celler, S. Cristoloveanu, F. Gamiz, J.G. Fossum, K. Izumi (Eds.), 12th International Symposium on Silicon-on-Insulator Technology and Devices, vol. 19, The Electrochemical Society, Pennington, NJ, USA, 2005, p. 357.
- [44] C. Moulin, D. Delprat, C. Maleville, W. McMillan, K. Birdwell, J. Payne, R. Brun, R. Moirin, in: Proc. IEEE International SOI Conf., 2005, p. 146.

- [45] A. Hamamatsu, H. Shibuya, Y. Oshina, S. Maeda, H. Nishiyama, M. Noguchi, in: 12th Asia Pacific Conference on NDT, Auckland, New Zealand, 2006.
- [46] S.R. Lange, B. Pinto, J. Fernandez, *Yield Manag. Solut.* 9 (1) (Winter 2007) 1.
- [47] D. Hendricks, J.Y. Jan, H. Dohse, A.D. Brodie, W.D. Meisburger, in: *SPIE Integrated Circuit Metrology, Inspection, Process Control IX*, vol. 2439, 1995, p. 174.
- [48] S. Abe, T. Sekiguchi, H. Nakano, M. Noguchi, *Hitachi Rev.* 55 (2) (2006) 73.
- [49] L.A. Cheema, L.J. Olmer, O.D. Patterson, S.S. Lopez, M.B. Burns, *Solid State Technol.* 46 (9) (2003) 57.
- [50] C. Saravanan, W. Yang, Z. Liu, M.F. Swisher, A. Tang, *MICRO* 22 (3) (April 2004) 29.
- [51] T. Bearda, P.W. Mertens, F. Holsteys, P. De Bisschop, R. Compen, A. van Meer, M.M. Heyns, *Jpn. J. Appl. Phys.* 44 (10) (2005) 7409.
- [52] S.-H. Yoo, J. Weygand, J. Scherer, L. Davis, B. Liu, K. Christenson, J. Butterbaugh, N. Narayanswami, *J. Vac. Sci. Technol. B* 19 (2) (2001) 344.
- [53] J.C. Stover, in: A. Diebold (Ed.), Chapter 21, *Handbook of Silicon Semiconductor Metrology*, Marcel Dekker, New York, NY, 2001.
- [54] G.W. Mulholland, M.K. Donnelly, C.R. Hagwood, S.R. Kukuck, V.A. Hackley, D.Y. Pui, *J. Res. Natl. Inst. Stand. Technol.* 111 (4) (2006) 257.
- [55] V. Sankaran, J.C. Stover, #99083800B-TR, SEMATECH, 1999.
- [56] G.W. Mulholland, T.A. Germer, J.C. Stover, in: *Proceedings of the Government Microcircuits Applications and Critical Technologies Conference*, Tampa, Florida, 2003.
- [57] H.J. Harrington, G.D. Hoffherr, R.P. Reid, *Statistical Analysis Simplified: The Easy-to-Understand Guide to SPC and Data Analysis*, McGraw-Hill, New York, NY, 1998.
- [58] K.L. Mittal, *Particles in Gases and Liquids 1: Detection, Characterization, and Control*, Plenum Press, New York, 1989.
- [59] E. Terrell, in: M. Balazs (Ed.), 24th *Proceedings of Semiconductor Pure Water and Chemicals Conference (SPWCC)*, 2005, p. 91. Santa Clara, CA.
- [60] R.G. Knollenberg, D.L. Veal, in: *Presented at the 37th Annual Meeting of the Institute of Environmental Sciences and Technology (ZEST)*, 1991, p. 751.
- [61] E. Terrell, J. Gromala, D. Beal, *Understanding Liquid Particle Counters*, Particle Measuring Systems Inc., 2005.
- [62] T. Pietrykowski, J. Mitchell, *J. Ultrapure Water* (May 2009).
- [63] D. Rodier, in: *Ultrapure Water Conference*, December 2014.
- [64] P. Scardina, R.D. Letterman, M. Edwards, *J. Am. Water Works Assoc.* 98 (7) (2006) 97.
- [65] A. Dey, in: N.N. Li, A.G. Fane, W.S. Winston Ho, T. Matsuura (Eds.), *Advanced Membrane Technology and Applications*, John Wiley & Sons, Hoboken, NY, 2008, p. 371.
- [66] D. Rodier, in: *Surface Preparation and Cleaning Conference (SPCC)*, March 28, 2017, pp. 03–07.
- [67] J. Mitchell, in: *Presented at the 44th Annual Meeting of the Institute of Environmental Sciences and Technology (ZEST)*, 1998, p. 751.
- [68] B.A. Knollenberg, *DataTech*, September 1999, p. 3.
- [69] *ISO 21501-4 Institute of Environmental Sciences & Technology (IEST)*, 2015.
- [70] T. Harrison, B. Latimer, *Cleanrooms* 22 (7) (2008) 36.
- [71] S. Libman, A. Neuber, B. Latimer, S. Schoen, D. Sinha, D. Blackford, in: *Ultrapure Water (UPW) Conference*, Portland, OR, 2009.
- [72] J. Gromala, D. Beal, *Cleanroom Technology*, April 2004, p. 21.
- [73] D. Rodier, in: *SEMI Technology Symposium (STS) Contamination Free Manufacturing*, Session 5.2, February 2017.
- [74] D.B. Blackford, K.J. Belling, G.J. Sem, *J. Environ. Sci.* 30 (4) (1987) 43.

- [75] D.B. Blackford, G.J. Sem, T. Kerrick, in: *Proceedings of the International Conference of Particle Detection, Metrology and Control*, 1990, p. 546. Arlington, VA.
- [76] D.B. Blackford, F. Zarrin, G. Sem, T. Kerrick, in: *Proceedings of the Microcontamination Conference and Exposition*, Canon Communications, San Jose, CA, 1991, p. 39.
- [77] ISO 14644-1 and ISO 14644-2, Institute of Environmental Sciences & Technology (IEST), 2015.
- [78] S. Kochevar, *Cleanroom Technology*, November 2005.
- [79] ISO 14644-8, Institute of Environmental Sciences & Technology (IEST), 2013.
- [80] T. Fujimura, K. Takeda, T. Nonaka, *Developments in surface contamination and cleaning*, in: second ed., in: K. Rajiv, K.L. Mittal (Eds.), *Fundamentals and Applied Aspects*, vol. 1, Elsevier, Amsterdam, 2015, p. 197.
- [81] M.L. Kwan, C. Muller, S.B. Tan, R. Thomas, in: *IEEE/SEMI Advanced Semiconductor Manufacturing Conference (ASMC)*, 2006, p. 383.
- [82] I.-K. Lin, H. Bai, C.-C. Liu, B.-J. Wu, *Sep. Sci. Technol.* 43 (2008) 842.
- [83] M. Polen, B. Shade, *Cleanrooms* 22 (7) (2008) 28.
- [84] S. Kochevar, T. Pietrykowski, D. Rodier, in: *SPIE Optical System Contamination: Effects, Measurements, and Control*, vol. 8492, 2012, p. 0H.
- [85] N. Gambino, B. Rollinger, D. Hudgins, R.S. Abhari, *J. Micro Nanolithogr. MEMS MOEMS* 14 (3) (2015) 034002.
- [86] R. Zhou, C. Edwards, C. Bryniarski, G. Popescu, L.L. Goddard, in: *SPIE Metrology, Inspection, and Process Control for Microlithography XXIX*, vol. 9424, 2015, p. 241.
- [87] R. Bonam, H.-Y. Tien, A. Chou, L. Meli, S. Halle, I. Wu, X. Huang, C. Lei, C. Kuan, F. Wang, D. Corliss, W. Fang, J. Jau, Z.J. Qi, K. Badger, C. Turley, J. Rankin, in: *SPIE Extreme Ultraviolet (EUV) Lithography VII*, vol. 9776, 2016, p. 1C.
- [88] B.D. Bunday, M. Mukhtar, K. Quoi, B. Thiel, M. Malloy, in: *SPIE Metrology, Inspection, and Process Control for Microlithography XXIX*, vol. 9424, 2015, p. 0J.
- [89] T. Kemen, T. Garbowski, D. Zeidler, in: *Photomask and Next-Generation Lithography Mask Technology XXII*, vol. 9658, 2015, p. 07.
- [90] M. Mostafavi, R.E. Diaz, *Nat. Sci. Rep.* 6 (2016) 26172.
- [91] B.M. Barnes, H. Zhou, M.A. Henn, M.Y. Sohn, R.M. Silver, in: *SPIE Modeling Aspects in Optical Metrology VI*, vol. 10330, 2017, p. 0W.
- [92] P. van der Walle, E. Kramer, J.C. van der Donck, W. Mulckhuyse, L. Nijsten, F.B. Arango, A. de Jong, E. van Zeijl, H.E. Spruit, J.H. van den Berg, G. Nanda, in: *SPIE Optical Measurement Systems for Industrial Inspection X*, vol. 10329, 2017, p. 4N.
- [93] K. Tachibana, M. Masaki, T. Satoru, T. Kiyoshi, in: *International Conference on Leading Edge Manufacturing in 21st Century (LEM21)*, vol. 8, 2015, 1507–1.
- [94] A. Sengupta, *Surface Preparation and Cleaning Conference (SPCC)*, March 28, 2017, pp. 01–07.
- [95] S. Libman, D. Wilcox, B. Zerfas, *ECS Trans.* 69 (8) (2015) 17.
- [96] A. Rastegar, M. Samayoa, M. House, H. Kurtuldu, S.K. Eah, L. Morse, J. Harris-Jones, in: *SPIE Extreme Ultraviolet (EUV) Lithography V*, vol. 9048, 2014, p. 0P.